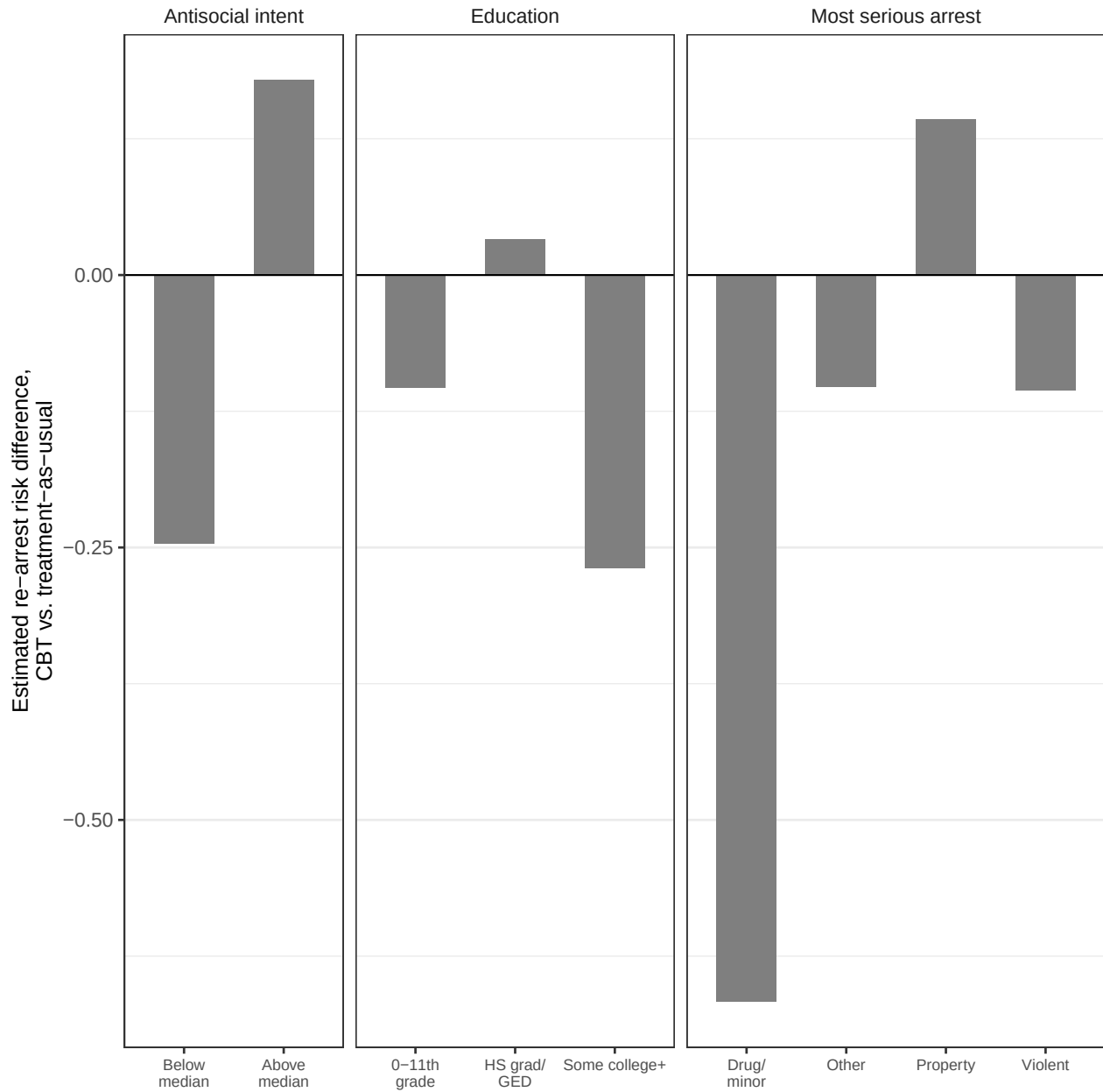


Results for DDDM paper

1 Subgroup analysis

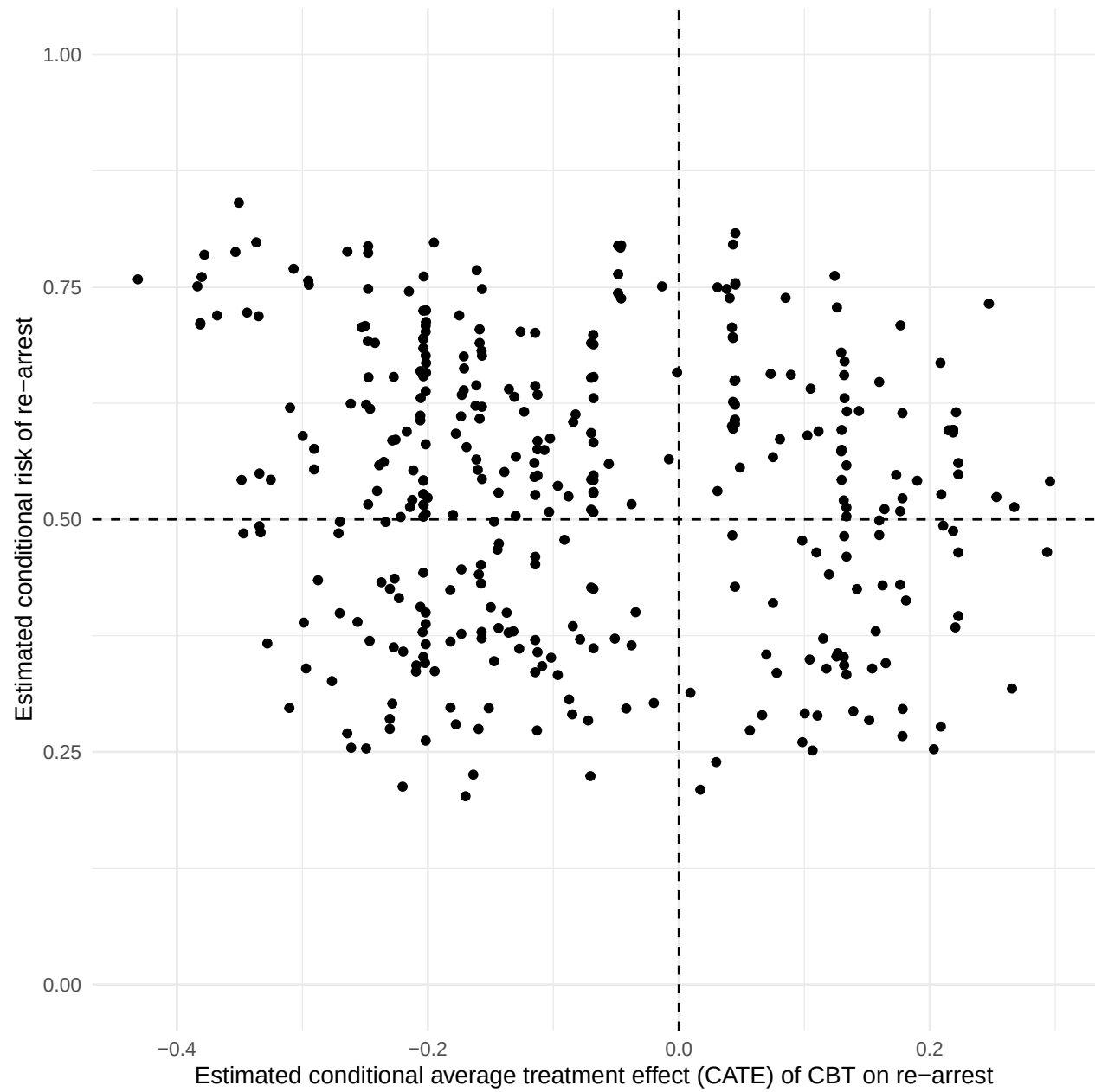


2 Risk score

2.1 SuperLearner

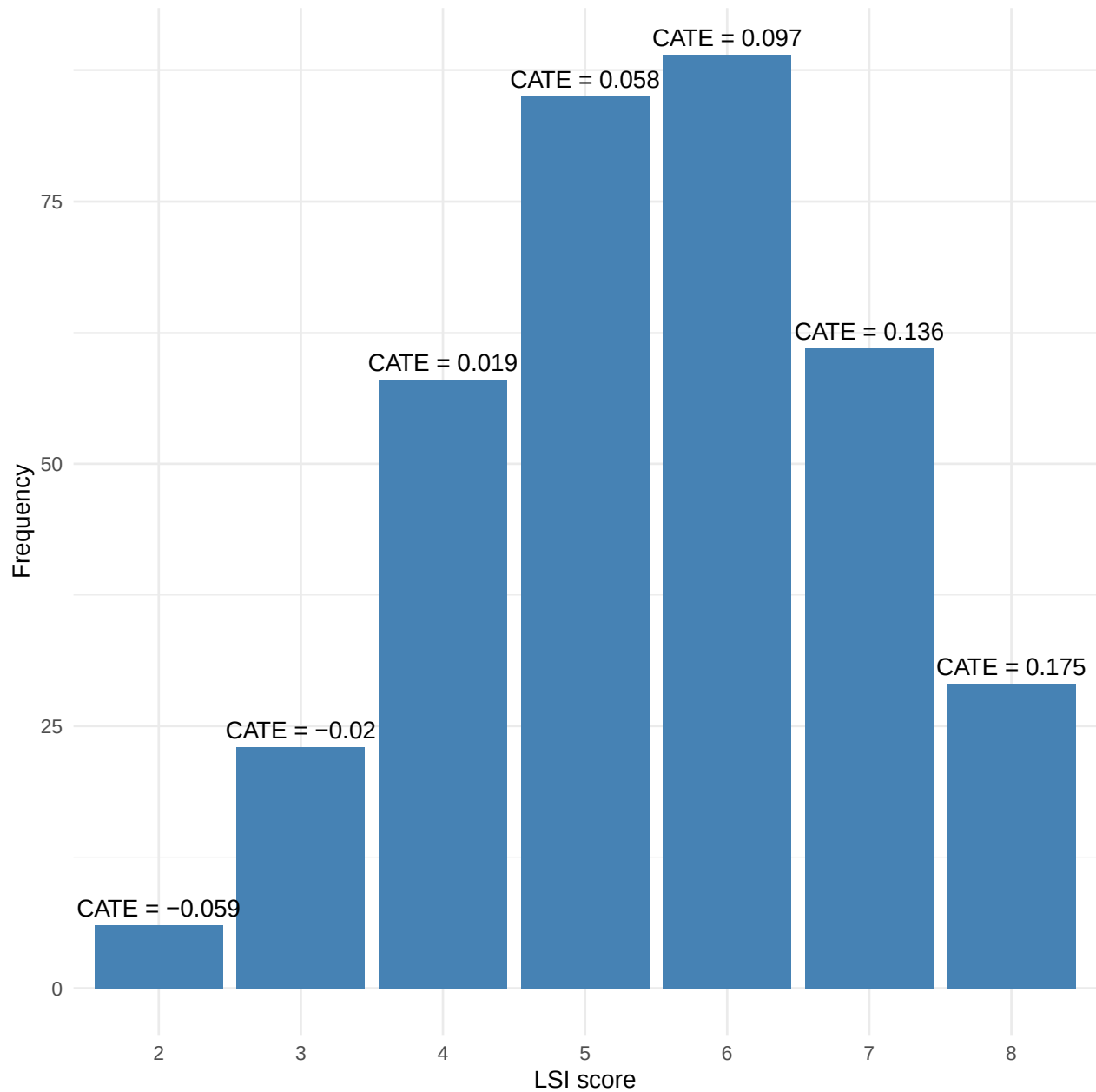
```
Call:  
SuperLearner(Y = Y0, X = W0, family = "binomial", SL.library = blip.SL.library)
```

	Risk	Coef
SL.blip.W_Antisocial_IntentTotal_All	0.2520562	0.00000000
SL.blip.W_Arrest_MostSerious10.4_All	0.2466882	0.16526971
SL.blip.W_Arrest_NumUnder16_10.2_All	0.2498961	0.00000000
SL.blip.W_CCSTotal_All	0.2531656	0.00000000
SL.blip.W_CSITotal_All	0.2534063	0.00000000
SL.blip.W_Dem_Age1.2_All	0.2553337	0.00000000
SL.blip.W_Dem_Gender1.1_All	0.2472176	0.02669168
SL.blip.W_Dem_Hispanic1.3_All	0.2531224	0.00000000
SL.blip.W_Dem_Race_Cat_All	0.2537314	0.00000000
SL.blip.W_DERSTotal_All	0.2531006	0.00000000
SL.blip.W_EmploymentStatus3.2_All	0.2542711	0.00000000
SL.blip.W_HighestGrade3.1_All	0.2570827	0.00000000
SL.blip.W_IdentityTotal_All	0.2514698	0.00000000
SL.blip.W_LSIR_TotalScore15.9_All	0.2315625	0.44272032
SL.blip.W_MCAATotal_All	0.2533096	0.00000000
SL.blip.W_Most_Serious_All	0.2474640	0.00000000
SL.blip.W_NASTotal_All	0.2531715	0.00000000
SL.blip.W_Num_Adult_Prior_Conv_All	0.2466935	0.00000000
SL.blip.W_SCSTotal_All	0.2491616	0.00000000
SL.blip.W_SES_Ladder3.4R_All	0.2509064	0.00000000
SL.blip.W_SPSITotal_All	0.2499272	0.00000000
SL.blip.W_Substance_Abuse_All	0.2515979	0.00000000
SL.blip.W_TX_TMQTotal_All	0.2525692	0.00000000
SL.mean_All	0.2512512	0.00000000
SL.earth_All	0.3273059	0.00000000
SL.rpart_All	0.3259313	0.00000000
SL.glm_All	0.2633842	0.00000000
SL.randomForest_All	0.2445713	0.00000000
SL.bayesglm_All	0.2485738	0.36531829
SL.stepAIC_All	0.2587003	0.00000000
[1]	-0.1362474	



2.2 LSI

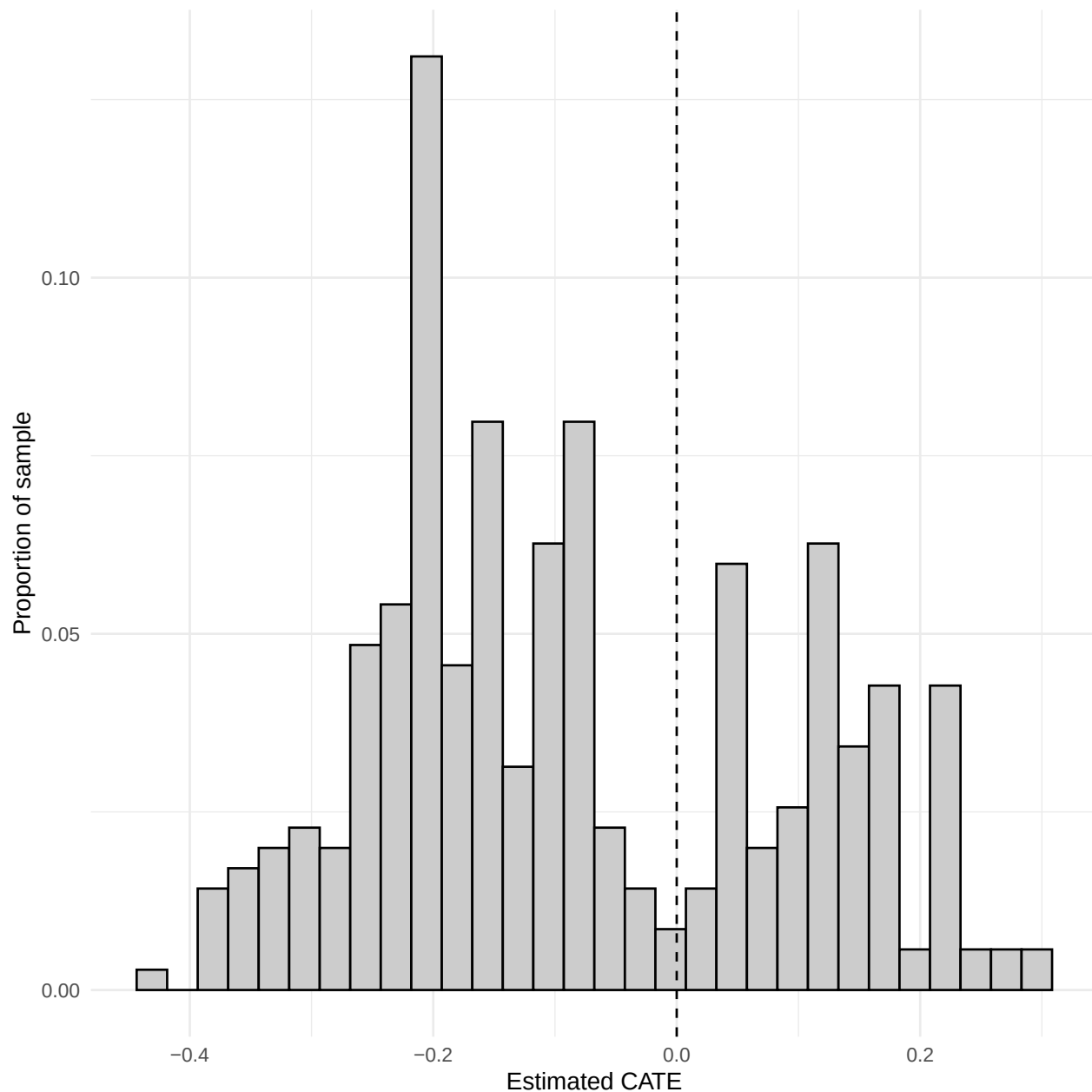
[1] 1



3 Optimal rule

SL.blip.W_Antisocial_IntentTotal	SL.blip.W_Antisocial_IntentTotal_subgrp
0.21955452	0.40499114
SL.blip.W_Arrest_MostSerious10.4	SL.blip.W_Arrest_NumUnder16_10.2
0.14437468	0.00000000
SL.blip.W_CCSTotal	SL.blip.W_CSITotal
0.00000000	0.00000000
SL.blip.W_Dem_Age1.2	SL.blip.W_Dem_Gender1.1
0.00000000	0.00000000
SL.blip.W_Dem_Hispanic1.3	SL.blip.W_Dem_Race_Cat
0.00000000	0.00000000

SL.blip.W_DERSTotal	0.00000000	SL.blip.W_EmploymentStatus3.2	0.00000000
SL.blip.W_HighestGrade3.1	0.01379205	SL.blip.W_IdentityTotal	0.00000000
SL.blip.W_LSIR_TotalScore15.9	0.00000000	SL.blip.W_MCAATotal	0.00000000
SL.blip.W_Most_Serious	0.00000000	SL.blip.W_NASTotal	0.00000000
SL.blip.W_Num_Adult_Prior_Conv	0.00000000	SL.blip.W_SCSTotal	0.00000000
SL.blip.W_SES_Ladder3.4R	0.00000000	SL.blip.W_SPSITotal	0.00000000
SL.blip.W_Substance_Abuse	0.00000000	SL.blip.W_TX_TMQTotal	0.00000000
SL.mean	0.00000000	SL.earth	0.21728760
SL.rpart	0.00000000	SL.glm	0.00000000
SL.randomForest	0.00000000	SL.bayesglm	0.00000000
SL.stepAIC	0.00000000		



```

0      1
114 237

0      1
114 237

0      1
0.3247863 0.6752137
% latex table generated in R 4.4.2 by xtable 1.8-4 package
% Fri Oct 2 10:18:43 2026
\begin{table}[ht]
\centering
\begin{tabular}{rlll}
\toprule
& level & TAU & CBT \\\end{pre>
```

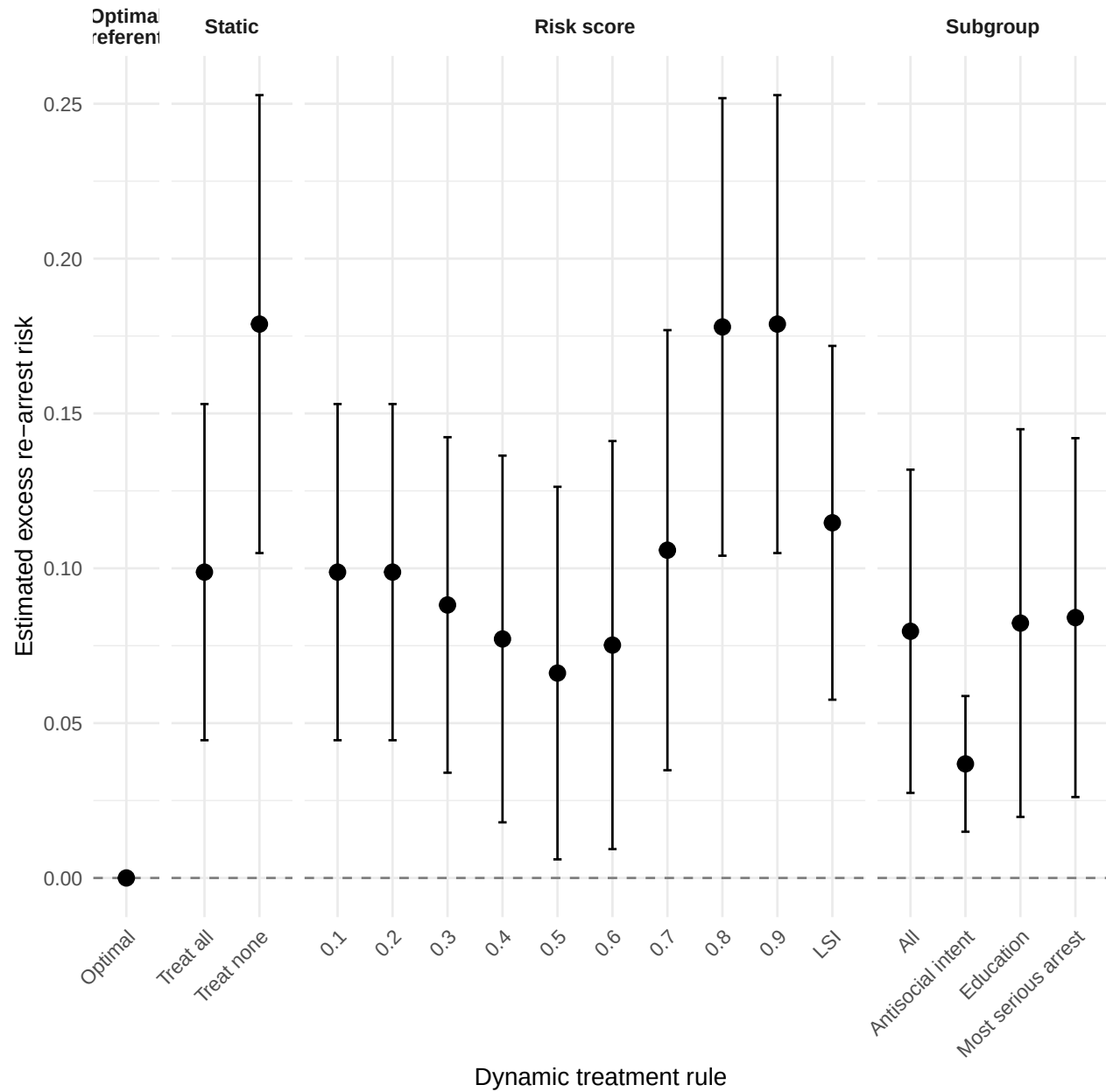
```

\midrule
n & & 114 & 237 \\
Dem.Age1.2..mean..SD.. & & 37.75 (11.73) & 38.19 (10.63) \\
Dem.Gender1.1.... & Male & 96 (84.2) & 200 (84.4) \\
X & Female & 18 (15.8) & 37 (15.6) \\
Dem.Hispanic1.3.... & No & 89 (78.1) & 198 (83.5) \\
X.1 & Yes & 25 (21.9) & 39 (16.5) \\
Dem.Race.Cat.... & All other groups & 23 (20.2) & 53 (22.4) \\
X.2 & Black & 37 (32.5) & 96 (40.5) \\
X.3 & White & 54 (47.4) & 88 (37.1) \\
HighestGrade3.1.... & 0-11th grade & 31 (27.2) & 76 (32.1) \\
X.4 & High school graduate/GED & 60 (52.6) & 93 (39.2) \\
X.5 & Some college/grad/postgrad & 23 (20.2) & 68 (28.7) \\
EmploymentStatus3.2.... & Part time/full time & 6 (5.3) & 19 (8.0) \\
X.6 & Unemployed/retired & 108 (94.7) & 218 (92.0) \\
SES.Ladder3.4R.... & 1 & 20 (17.5) & 33 (13.9) \\
X.7 & 2 & 26 (22.8) & 13 (5.5) \\
X.8 & 3 & 10 (8.8) & 37 (15.6) \\
X.9 & 4 & 13 (11.4) & 36 (15.2) \\
X.10 & 5 & 27 (23.7) & 56 (23.6) \\
X.11 & 6/7/8/9 & 18 (15.8) & 62 (26.2) \\
LSIR.TotalScore15.9..mean..SD.. & & 5.86 (1.32) & 5.33 (1.44) \\
Most.Serious.... & minor or drug & 7 (6.1) & 13 (5.5) \\
X.12 & none & 25 (21.9) & 44 (18.6) \\
X.13 & property & 30 (26.3) & 59 (24.9) \\
X.14 & violent or sex or other person & 52 (45.6) & 121 (51.1) \\
Arrest.MostSerious10.4.... & Drug crimes and minor crimes & 1 (0.9) & 12 (5.1) \\
X.15 & Other person & 14 (12.3) & 22 (9.3) \\
X.16 & Property crimes & 17 (14.9) & 30 (12.7) \\
X.17 & Violent & 82 (71.9) & 173 (73.0) \\
Arrest.NumUnder16.10.2.... & Never & 49 (43.0) & 132 (55.7) \\
X.18 & One or more times & 65 (57.0) & 105 (44.3) \\
Num.Adult.Prior.Conv..mean..SD.. & & 5.63 (6.26) & 5.22 (5.15) \\
CSITotal..mean..SD.. & & 37.46 (10.63) & 31.78 (11.82) \\
Substance.Abuse.... & Neither too much nor cut down & 28 (24.6) & 84 (35.4) \\
X.19 & Too much and cut down & 57 (50.0) & 98 (41.4) \\
X.20 & Too much or cut down & 29 (25.4) & 55 (23.2) \\
SCSTotal..mean..SD.. & & 24.20 (7.87) & 31.01 (9.30) \\
CCSTotal..mean..SD.. & & 2.40 (0.39) & 2.26 (0.38) \\
MCAATotal..mean..SD.. & & 14.97 (2.79) & 13.20 (3.30) \\
DERSTotal..mean..SD.. & & 27.97 (12.35) & 22.21 (14.67) \\
NASTotal..mean..SD.. & & 6.12 (4.57) & 4.38 (4.50) \\
SPSITotal..mean..SD.. & & 27.28 (4.50) & 29.16 (4.73) \\
IdentityTotal..mean..SD.. & & 4.73 (2.93) & 2.61 (2.50) \\
TX.TMQTotal..mean..SD.. & & 20.69 (7.40) & 21.96 (7.74) \\
Antisocial.IntentTotal..mean..SD.. & & 6.20 (1.37) & 2.70 (1.45) \\
\bottomrule
\end{tabular}
\caption{Baseline characteristics stratified by optimal treatment recommendation}
\label{tab:table1_odtr}
\end{table}

```

4 Values and contrasts

```
[1] 0.3494596 0.4141784 0.2847409
```



```
...1          Psi      CI1      CI2
treatall      0.09876282 0.044477956 0.1530477
treatnone     0.17885586 0.104913074 0.2527986
rho_0.1       0.09876282 0.044477956 0.1530477
rho_0.2       0.09876282 0.044477956 0.1530477
rho_0.3       0.08815549 0.033982170 0.1423288
rho_0.4       0.07717431 0.017962672 0.1363859
rho_0.5       0.06615909 0.006001780 0.1263164
```

rho_0.6	0.07520192	0.009320378	0.1410835
rho_0.7	0.10584285	0.034784315	0.1769014
rho_0.8	0.17793727	0.104060405	0.2518141
rho_0.9	0.17885586	0.104913074	0.2527986
dtr_LSI	0.11469300	0.057567121	0.1718189
subgroupdtr_W_HighestGrade3.1	0.08230121	0.019704316	0.1448981
subgroupdtr_W_Arrest_MostSerious10.4	0.08407067	0.026123110	0.1420182
subgroupdtr_W_Antisocial_IntentTotal	0.03683908	0.014928455	0.0587497
subgroupdtr_W_all	0.07966192	0.027468472	0.1318554

	dtr_name	rule_group
...1	Optimal	Optimal\n(referent)
treatall	Treat all	Static
treatnone	Treat none	Static
rho_0.1	0.1	Risk score
rho_0.2	0.2	Risk score
rho_0.3	0.3	Risk score
rho_0.4	0.4	Risk score
rho_0.5	0.5	Risk score
rho_0.6	0.6	Risk score
rho_0.7	0.7	Risk score
rho_0.8	0.8	Risk score
rho_0.9	0.9	Risk score
dtr_LSI	LSI	Risk score
subgroupdtr_W_HighestGrade3.1	Education	Subgroup
subgroupdtr_W_Arrest_MostSerious10.4	Most serious arrest	Subgroup
subgroupdtr_W_Antisocial_IntentTotal	Antisocial intent	Subgroup
subgroupdtr_W_all	All	Subgroup

```

\begin{table}[ht]
\centering
\begin{tabular}{lllc}
\toprule
Rule group & Treatment rule & Risk difference & 95\% CI \\
\midrule
1 & Optimal & 0.00 & (0.000, 0.000) \\
2 & Treat all & 0.10 & (0.044, 0.153) \\
& & & & Treat none & 0.18 & (0.105, 0.253) \\
3 & 0.1 & 0.10 & (0.044, 0.153) \\
& & & & 0.2 & 0.10 & (0.044, 0.153) \\
& & & & 0.3 & 0.09 & (0.034, 0.142) \\
& & & & 0.4 & 0.08 & (0.018, 0.136) \\
& & & & 0.5 & 0.07 & (0.006, 0.126) \\
& & & & 0.6 & 0.07 & (0.009, 0.141) \\
& & & & 0.7 & 0.11 & (0.035, 0.177) \\
& & & & 0.8 & 0.18 & (0.104, 0.252) \\
& & & & 0.9 & 0.18 & (0.105, 0.253) \\
& & & & LSI & 0.12 & (0.058, 0.172) \\
4 & Education & 0.08 & (0.020, 0.145) \\
& & & & Most serious arrest & 0.08 & (0.026, 0.142) \\
& & & & Antisocial intent & 0.04 & (0.015, 0.059) \\
& & & & All & 0.08 & (0.027, 0.132) \\
\bottomrule

```

```

\end{tabular}
\caption{Estimated risk difference in 2-year recidivism relative to the optimal treatment rule.}
\label{tab:contrast_results}
\end{table}

```